

PLUGIN Platform

Future Enterprise Features & Research Roadmap

This document outlines the strategic roadmap for high-impact, software-only features designed to elevate the PLUGIN mobile app and backend into a multi-million dollar enterprise-grade EV platform. These features rely heavily on the existing Spring Boot and MongoDB architecture.

1. Core Driver Convenience & Utility

Smart Route Planner (Battery-Aware Navigation)

Calculates entire journeys for long road trips, automatically adding PLUGIN stations as mandatory charging waypoints to prevent range anxiety.

Implementation: Uses userVehicles data (battery capacity, EV model) to calculate range and integrate waypoints via Spring Boot.

Zero-Friction Pre-Paid Wallet

An in-app wallet where users preload funds to avoid heavy credit card gateway fees on individual small transactions.

Implementation: Combines with QR Code scanning on physical hardware to auto-deduct funds and start charging instantly.

Weather-Aware Charging Speed Predictor

Adjusts user expectations by warning them if extreme temperatures (hot or freezing) will slow down their DC fast charging.

Implementation: Spring Boot integrates with a free weather API (like OpenWeatherMap) to adjust ETA calculations and trigger mobile UI warnings.

2. Station Flow & Queue Optimization

The "Virtual Queue" (Waitlist Notifications)

Eliminates physical waitlines by allowing users to join a virtual queue and receive push notifications when a charger becomes available.

Implementation: Backend detects session completion in `chargingSessions` and pings the next user in the waitlist.

Predictive Availability Forecasting ("Crystal Ball")

Shows a historical usage graph for stations (e.g., "Usually busy between 5 PM and 7 PM") so users can plan ahead.

Implementation: Spring Boot runs an aggregation query on the MongoDB `chargingSessions` collection to calculate historical average usage.

Peer-to-Peer "Hand-Off" Coordination

Allows the person currently charging to "ping" the next person in the virtual queue (e.g., "Leaving in 5 mins").

Implementation: Uses the existing notifications collection to alert the next user, fostering a community-driven flow.

3. Cost Management & Hardware Protection

Smart Scheduling (The Cost Saver)

Allows users parked overnight to schedule charging during off-peak hours (e.g., 2:00 AM) to save money.

Implementation: Spring Boot checks local electricity pricing APIs and schedules `@Scheduled` tasks to start the charger automatically later.

Anti-Hogging Idle Fees

Automated warning and penalty system for cars that sit idle at 100% battery, blocking the physical hardware.

Implementation: Backend starts a high-priority timer. If it hits zero, the billing engine auto-charges the user's wallet per minute.

Battery Health Auto-Stop ("80% Rule")

Users can set the charger to auto-stop at 80% to protect battery health and free up the physical station much faster.

Implementation: Spring Boot calculates required kWh based on userVehicles and terminates the session early via scheduled tasks.

The "Clean Bay" Loyalty Reward

Gamifies station etiquette. Users snap a photo showing they holstered the cable correctly to earn a micro-reward, preventing cable damage.

Implementation: Photo verification tied to automatic in-app wallet credits.

4. Environmental & Community Impact

"Eco-Charge" (Grid-Aware Renewable Charging)

Automatically aligns charging sessions with periods when the local electrical grid is using the highest percentage of renewable energy.

Implementation: Integrates with APIs like Electricity Maps to optimize the exact start time of an overnight session.

Wait-Time Commerce & Amenity Hub

Partners with local coffee shops. Users get a temporary discount QR code while their car charges.

Implementation: When a session starts in MongoDB, the React Native app generates a time-limited 10% off coupon for local businesses.

Community Micro-Tips

Allows users to leave short tips for specific physical charging points (e.g., "Pro Tip: Lift the handle up slightly while it connects!").

Implementation: Adds a simple quirkTips array to the MongoDB chargingPoints collection.